Sahaj Bhadja

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Summary

Software engineer with 2+ years building scalable customer-facing systems, distributed workflows, and cross-platform applications. Experienced in Java, Python, C++, TypeScript, object-oriented design, algorithms, AWS, relational databases, automated testing, performance optimization, and Agile delivery.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Reduced user-facing dashboard latency from 1.2 seconds to 350 milliseconds for 500+ users by profiling Python API request paths and tuning Azure SQL queries, indexes, and secure access patterns.
- Built FastAPI services and fault-tolerant asynchronous LLM workflows for 5,000+ monthly jobs, using explicit states, retries, reconciliation, and failure isolation to triple throughput while sustaining 99%+ reliable production processing.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed Flask APIs and React implementation workflows supporting 150+ users and 10,000+ daily requests, integrating secure ingestion, authentication, AWS RDS persistence, status tracking, and instructor-facing product experiences.
- Designed concurrent processing for 1,000+ uploaded files with real-time job-state updates, automated validation, and fault-tolerant execution, reducing turnaround 60% while maintaining stable API performance during peak demand.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built reproducible simulation and benchmarking pipelines processing 100,000+ records, applying structured automated tests, instrumentation, and systematic debugging to improve long-horizon planning accuracy 18% across controlled evaluations.
- Participated in cross-functional collaborative design reviews and validation experiments, documenting technical decisions and tuning a performance-critical architecture that improved closed-loop stability 15% across repeated controlled simulation scenarios.

Aasma Technology Solutions **India**
Software Developer Engineer – SecureSync *August 2023 – July 2024*
- Maintained 20+ automated unit and integration suites using Jenkins, PyTest, and Mockito across Windows, Linux, and macOS, accelerating defect diagnosis and improving confidence in cross-platform production releases.
- Partnered with Microsoft and GitHub stakeholders on API integrations, then optimized filtering, pagination, indexing, and caching to improve client response times 30% and production data-pipeline efficiency 68%.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
- Built React Native clients for iOS and Android with Firebase persistence, real-time state synchronization, image storage, push notifications, and secure sessions across a customer platform handling 50,000+ records.
- Developed Flask APIs for bookings, pricing, and payments, delivering sub-200-millisecond responses and webhook-driven synchronization while improving confirmation speed 40% across customer-facing mobile and web experiences during peak demand.

Skills

Languages & Core Engineering: Java, Java 17, Python, C++, TypeScript, JavaScript, SQL, Bash, object-oriented design, data structures, algorithms, design patterns, performance optimization
Backend & Distributed Systems: Java services, REST APIs, Flask, FastAPI, Node.js, microservices, WebSockets, concurrency, multithreading, asynchronous workflows, real-time synchronization, caching, failure isolation
Data & Cloud: AWS, AWS RDS, S3, PostgreSQL, MySQL, SQL Server, Azure SQL, MongoDB, Firebase, relational databases, NoSQL, indexing, query optimization, secure storage
Quality & Delivery: Maven, Git, GitHub, CI/CD, Jenkins, GitHub Actions, Docker, PyTest, Mockito, unit testing, integration testing, regression testing, code review, debugging, monitoring, documentation, Agile, Scrum

Projects

Chat2SQL | Code | *Java 17, TypeScript, SQL, Docker, Maven*
- Built a cross-platform SQL development workspace supporting 10+ databases, combining Java 17 services, TypeScript, schema-aware query generation, validation, Docker, and Maven workflows for reliable data exploration and management.

CPU-Scheduling Simulator | Live | *C++, React, JavaScript, Algorithms*
- Implemented six scheduling algorithms with dynamic metrics and Gantt visualization across 1,000+ process combinations, reducing debugging time 20% through deterministic simulation, maintainable C++ logic, and systematic edge-case testing.

FinNexus | Live | *TypeScript, WebSockets, MongoDB, Kubernetes*
- Built a real-time platform supporting 50,000+ users and one million transactions, integrating secure APIs, WebSocket synchronization, advanced filtering, MongoDB persistence, and reliable scalable containerized production deployment.